Zixuan Chen

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Education

Carnegie Mellon University **Pittsburgh, PA, USA**
Master of Science in Information Networking - Advanced Study August 2025 - May 2027

- Coursework: 15-746 Storage Systems, 15-641 Computer Networks and the Internet, 15-513 Introduction to Computer Systems, 14-741 Introduction to Information Security

Shanghai Jiao Tong University **Shanghai, China**
Bachelor of Engineering in Information Security September 2018 - June 2022

- Coursework: Operating Systems, Computer Networks, Linux Kernel, Artificial Intelligence, Compiler, Database, etc

Work Experience

ByteDance **Beijing, China**
Software Engineer, Hybrid Cloud Computing Team July 2022 - April 2025

- **Scheduling System.** Maintained scheduling and resource management system for virtual machines in Hybrid Cloud Computing Team; designed and applied resource scheduling strategies for GPU and RDMA in consideration of intra-node and inter-node topology; developed resource reservation mechanism for VM migration and resize. Expanded capabilities of resource scheduler and controller to support more diverse product formations
- **GPU and HPC Instance Type.** Led the development of Nvidia L20 and A10 GPU instance types; participated in the development of BlueField 3 RDMA instance types; main tasks included adapted resource controller and scheduler in control plane, expanded the monitor agent, and conducted network communication benchmark using NCCL and LLM model inference benchmark evaluation using vllm
- **Reliability and Observability.** Participated in developing event-driven automatic failure handling system for virtual machines; designed and implemented event notification task using CloudEvent; orchestrated workflows for failure handling. Developed monitor agent plugins for VM status collection and live migration downtime using libvirt API

Projects

Disaggregated Pipeline Parallelism for LLM Serving
Carnegie Mellon University Catalyst Group, Advised by Professor Zhihao Jiao October 2025 - Present

- Built the framework of cost model for attention, projection, norm, MoE and collective communication (NCCL) kernels
- Developed profilers for compute kernels using Nvidia Nsight System, profilers for collective communication kernels using CUDA events, and profilers for tensor core utilization using Nvidia Nsight Compute

CloudFS, October 2025 - December 2025

- Designed and developed a fault-tolerant hybrid cloud file system that stores file metadata on local SSD and offloads file chunks to Amazon S3 object store
- Implemented segmentation level deduplication along with write-back caching to optimize cloud storage costs

CMUTube, November 2025 - December 2025

- Implemented the BBA-0 adaptive bitrate algorithm using open-source dash.js
- Enhanced BBA-0 and got an average 150.46 quality of experience (QoE) score, 3rd place among all the groups

ThemiX - A Timing-balanced BFT Protocol
SJTU Decentralized Computing Lab, Advised by Professor Shengyun Liu August 2021 - May 2022

- Proposed and implemented ThemiX BFT protocol using Golang. Applied optimistic responsiveness and bypassing coin optimization, enabling fast path feature and allowing the protocol proceed under actual network latency
- Deployed and Evaluated ThemiX protocol in a network with 100 nodes across the world on AWS EC2 platform
- Published in Proceedings of the 24th International Middleware Conference (Middleware ’23)

Skills

Programming Languages: C/C++, Python, Golang, Rust
Tech Skills: vllm, NCCL, PyTorch, Kubernetes, Kafka, KVM, qemu, libvirt, Prometheus, InfluxDB, Telegraf, AWS EC2
Areas of Interest: Machine Learning Systems, Cloud Computing, GPU Communication